LESSON PLAN

MOCKUP SQUADRON ENVIRONMENT (MSE) (24)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	21
Total No. of Lesson Plans	21	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Do's and don'ts of Electrostatic sensitive device handling
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Do’s and don’ts of Electrostatic sensitive device handling.
5. Specific Objectives	24.15.1.1 Trainee will be able to follow Electrostatic sensitive device handling practices in a workshop.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of proper ESD wrist strap attachment, mat grounding verification, and safe component placement on anti-static surfaces.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary risk posed by static electricity to electronic components?
(b) Have you ever observed a component fail immediately after being removed from its packaging?
(c) Can you identify three common materials in a workshop that generate static charges?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of Electrostatic Discharge (ESD) • Mechanism of charge generation in workshops • Consequences of ESD on sensitive devices • Importance of ESD control measures	L & Ds	PPT, IB	15 min
(b)	• Do's: Proper grounding techniques • Do's: Use of anti-static packaging • Do's: Handling procedures for components • Don'ts: Prohibited actions near sensitive devices	L & Ds	PPT, PDF	20 min
(c)	• Inspection of ESD equipment functionality • Storage requirements for sensitive devices • Emergency response to suspected ESD damage • Maintenance of ESD workstations	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain why touching a metal surface before handling a component is considered a 'Don't'.
(b) Describe the correct procedure for verifying that an ESD wrist strap is functioning properly.
(c) List two specific storage conditions required for Electrostatic Sensitive Devices.

12. Home Assignment and Follow Up	Prepare a brief report detailing the specific ESD precautions required when transitioning from this lesson to the next lesson plan (LP 02) regarding component installation.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	End of subject	Refer to the master syllabus reference list	-

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min